

# THE MASTER QUADRATIC $P_{G^*}$ AS A MATH-PHYSICS BRIDGE: $\chi_{-4}$ , THE TERNARY SUBSTRATE, AND THE FTD IDENTIFICATION

**ABSTRACT.** The companion paper [1] establishes that the Kronecker character  $\chi_{-4}$  of  $K = \mathbb{Q}(i)$  generates the entire algebraic-analytic identity ring of the lemniscatic constant  $G^* = \Gamma(1/4)/\Gamma(3/4)$  through four functorial projections (lattice, Chowla-Selberg, Hecke, Dirichlet), and that  $\chi_{-4}(n) = \text{Im}(i^n)$  with value set  $\{-1, 0, +1\}$ . This companion note draws the consequences of those results for the math-physics bridge of the Foundational Ternary Dynamics (FTD) framework [2], where the same set  $\{-1, 0, +1\}$  appears as the ternary voxel alphabet of a discrete physical substrate. Three structural observations are recorded, all explicitly tagged as synthesis rather than theorem: (i) the FTD voxel alphabet coincides as a set with the value set of  $\chi_{-4}$ , sourcing a candidate *Gaussian-integer-module ontology*; (ii) the Born rule  $\psi \mapsto |\psi|^2$  and the character  $\chi_{-4} : i^n \mapsto \text{Im}(i^n)$  are the same arrow — a real-coordinate projection of a complex object — applied at different scopes; (iii) the FTD identification  $(x_+, x_-) \approx (\alpha^{-1}, N_c)$  of the roots of the master quadratic  $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$  with the fine-structure constant and the number of QCD colours requires exactly one bit of physical input — the exponent-pair selector  $(2, 3)$  — which is over-determined by joint root-matching. All mathematical content is taken from [1]; this companion contributes only the bridge interpretation, presented under the project's [SYNTHESIS] tag.

## 1. SETTING AND SCOPE

We adopt the entire mathematical content of [1] without re-derivation. The constants in use are:

- $G^* = \Gamma(1/4)/\Gamma(3/4) \approx 2.9587$ ;
- $G_G = 1/\text{AGM}(1, \sqrt{2}) \approx 0.8346$ , with the bridge identity  $G^* = 2\sqrt{\pi} G_G$ ;
- $\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \rightarrow \{\pm 1\}$ , the unique non-trivial Dirichlet character mod 4;
- the master quadratic  $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ , with integer coefficient  $16 = |\text{Aut}_{\mathbb{C}}(E_{\text{lemn}})|^2$  and roots  $x_{\pm}$  satisfying Vieta's identities  $x_+ + x_- = 16G^{*2}$ ,  $x_+x_- = 16G^{*3}$ ,  $1/x_+ + 1/x_- = 1/G^*$ .

The mathematical paper [1] proves that  $\chi_{-4}$  admits the explicit form  $\chi_{-4}(n) = \text{Im}(i^n) = \sin(\pi n/2)$  and that its value set is

$$\{\chi_{-4}(n) : n \in \mathbb{Z}\} = \{i^2, 0, |i^2|\} = \{-1, 0, +1\}.$$

Numerical observation: the roots of  $P_{G^*}$  are  $x_+ \approx 137.036 \dots$  (matching the inverse fine-structure constant  $\alpha^{-1}$  to 1.26 ppm) and  $x_- \approx 3.024 \dots$  (matching the number of QCD colours  $N_c = 3$  to 0.8%). The Foundational Ternary Dynamics framework [2] posits the identification  $(x_+, x_-) = (\alpha^{-1}, N_c)$  as a strongly motivated conjecture.

This note is the math-physics bridge: it draws structural consequences of the  $\chi_{-4}$ -unification of [1] for the FTD identification, without proving the identification itself. All load-bearing claims are tagged [SYNTHESIS] or [STRONGLY MOTIVATED CONJECTURE] as appropriate; no new mathematical theorem is claimed in this note.

## 2. THE VALUE-SET COINCIDENCE AND THE GAUSSIAN-INTEGER-MODULE ONTOLOGY

*Observation* (Value-set coincidence). [Synthesis.] Three independent starting points terminate at the same three-element set:

- (1) the FTD physical primitive — each lattice site  $v$  takes a state  $s_v \in \{-1, 0, +1\}$  encoding *negative state, void, positive state* [2];

- (2) the value set of the unique non-trivial real Dirichlet character of conductor 4 [1, §16];
- (3) the closure of the imaginary-unit orbit  $\{i^n : n \in \mathbb{Z}\}$  under the imaginary-part map:  $\{\text{Im}(i^0), \text{Im}(i^1), \text{Im}(i^2), \text{Im}(i^3)\} = \{0, 1, 0, -1\}$ .

That these three sets coincide is a set-level identity, not yet a *functorial* identification. Whether the FTD voxel alphabet *arises from* the character  $\chi_{-4}$ , or only happens to share its value set, is a physical-modelling question that lies outside the scope of the mathematical paper.

*Remark* (Candidate ontology:  $\mathbb{Z}[i]$ . -substrate realism) [**Synthesis.**] The value-set coincidence of Observation 2 suggests a working ontology in which the empirical universe is the real-coordinate readout of a  $\mathbb{Z}[i]$ -module structure. Layering the established FTD framework against this ontology:

- voxels are positions in a  $\mathbb{Z}[i]^d$ -style lattice ( $d = 3$  in current FTD);
- voxel states  $s_v \in \{-1, 0, +1\}$  are the values  $\text{Im}(i^\bullet)$  readouts of the unit group  $\mathbb{Z}[i]^\times$ ;
- the flux field  $J \in \mathbb{R}^3$  is the continuous real part of an underlying complex object;
- time is the order parameter of the cyclic action  $\langle i \rangle$  at long wavelength;
- the mass-length calibration  $a_{\text{phys}} \equiv \ell_P$  is the choice of unit on the  $\mathbb{Z}[i]$ -module;
- dimensionless physical constants ( $\alpha$ ,  $N_c$ , mass ratios) are intrinsic invariants of the  $\mathbb{Z}[i]$ -module structure.

The ontological position is most accurately described as *Eleatic structural realism, specialized to the Gaussian-integer module*: Eleatic because the substrate is a single algebraic object; structural realist in the sense of Ladyman–Ross [4], where the entities are nodes in the structural network rather than independent objects; specialized because the particular structure is the maximal order of  $\mathbb{Q}(i)$  rather than an arbitrary mathematical structure. This is a sharper position than Tegmark’s broad mathematical-universe hypothesis [3]: not “all mathematical structures are physical” but “the physical is the  $\mathbb{Z}[i]$ -module slice of the mathematical.” Wigner’s “unreasonable effectiveness” [5] becomes, in this picture, eminently reasonable.

*Remark* (Falsifiability). [**Synthesis.**] The  $\mathbb{Z}[i]$ -substrate ontology is empirically refutable: if experiment ever produced an observable that requires a  $\mathbb{Z}[\rho]$ -module structure (Eisenstein, sixfold) that cannot be embedded in a  $\mathbb{Z}[i]$  ambient, or that requires  $\mathbb{Z}[\sqrt{-2}]$  or  $\mathbb{Z}[\sqrt{-7}]$  (other class-number-1 quadratic orders), the framework breaks. The position has therefore the falsifiability profile of a substantive empirical hypothesis, not a metaphysical claim.

### 3. THE BORN RULE AND $\chi_{-4}$ AS THE SAME ARROW

The deepest single structural connection that the  $\chi_{-4}$ -unification of [1] makes available is the identification of two distinct operations as instances of the same arrow.

*Observation* (Born rule and  $\chi_{-4}$  as the same arrow). [**Synthesis.**] The Born rule of quantum mechanics and the character  $\chi_{-4}$  are formally the same projection applied to different complex objects:

- **Born rule:**  $\psi \in \mathbb{C} \mapsto |\psi|^2 \in \mathbb{R}_{\geq 0}$ , projecting a complex amplitude onto its real modulus.
- **Character**  $\chi_{-4}$ :  $i^n \in \langle i \rangle \subset \mathbb{C} \mapsto \text{Im}(i^n) \in \{-1, 0, +1\}$ , projecting a complex unit onto its signed imaginary coordinate.

Both are functions of the form (complex structure)  $\xrightarrow{\text{(single real coordinate)}} \mathbb{R}$ . The Born rule reads the modulus;  $\chi_{-4}$  reads the imaginary part. They are dual projections from the same complex structure, both producing ternary or near-ternary outcomes precisely because the imaginary-part projection of the cyclic group  $\langle i \rangle$  takes exactly three distinct values.

*Remark* (Reflexive projection at three scopes). [**Synthesis.**] Under the FTD framework’s identification of the voxel state field with the substrate’s reflexive projection [2], the same arrow applies

at three nested scopes:

$$\underbrace{\text{reflexive projection of substrate}}_{\text{FTD voxel state} = \text{Im}(i^\bullet)} \sim \underbrace{\text{measurement event of amplitude}}_{\text{Born rule outcome} = |\psi|^2} \sim \underbrace{\text{imaginary-part trace of unit orbit}}_{\chi_{-4}(n) = \text{Im}(i^n)}.$$

The first and third are mathematically equivalent (the FTD voxel state field, viewed as a function on lattice sites with values in  $\mathbb{Z}[i]^\times \cup \{0\}$  read off through the imaginary-part map, is literally the character  $\chi_{-4}$  on the angular index). The second is conjectured-structurally equivalent to the first via the FTD reflexivity ontology; this is the load-bearing physical conjecture and is tagged [STRONGLY MOTIVATED CONJECTURE] accordingly.

#### 4. THE MASTER QUADRATIC AS THE BRIDGE

The FTD bridge from  $\chi_{-4}$ -derived mathematics to physical observables runs through the master quadratic  $P_{G^*}$ . We recall (from [1, Theorem 16.2]) that  $\chi_{-4}$  generates the integer coefficient  $16 = |\mathbb{Z}[i]^\times|^2$  (lattice projection) and the constant  $G^*$  (Chowla–Selberg projection), but does *not* generate the specific polynomial form  $x^2 - 16G^{*2}x + 16G^{*3}$  from classical CM/Weber/Hecke machinery (the three negative tests of [1, §16.5]).

The exponent pair  $(2, 3)$  on  $G^*$  in the polynomial coefficients is nevertheless *uniquely picked out* by joint root-matching:

*Observation* (Joint-matching uniqueness, restated from [1, §16.6]). **[Synthesis.]** Among polynomial forms  $y^2 - 16R^p y + 16R^q$  at  $R = G^*$  with  $(p, q) \in \mathbb{Z}^2$ ,  $|p|, |q| \leq 5$ , only the pair  $(p, q) = (2, 3)$  produces an ordered root pair  $(y_+, y_-)$  that simultaneously matches the FTD pair  $(\alpha^{-1}, N_c)$  to sub-percent precision. The match precision at  $(2, 3)$  is 1.26 ppm for  $y_+ = 137.036$  and 0.8% for  $y_- = 3.024$ . Other exponent pairs match one root but not both, or neither.

*Remark* (Information-theoretic minimality of the FTD bridge). **[Synthesis.]** The FTD math-physics bridge requires exactly *one bit* of input from the physical side — the exponent-pair selector  $(p, q) \in \{(2, 3), (3, 4), (4, 5), \dots, (1, 2), (2, 4), \dots\}$  — and that bit is over-determined by joint matching to two distinct physical constants  $(\alpha^{-1}, N_c)$ . Equivalently: the FTD master quadratic is the unique polynomial of the form  $y^2 - 16R^p y + 16R^q$  at  $R = G^*$  whose ordered root pair  $(y_+, y_-)$  jointly matches  $(\alpha^{-1}, N_c)$  to sub-percent precision.

This is not a derivation of the bridge — the joint match itself is the input that selects  $(p, q)$  — but it is the cleanest available *minimality statement*: the math-physics bridge in the FTD framework requires no parameters beyond the choice of one exponent pair, and the choice is over-determined (hence robust) against the joint match to two distinct physical constants.

*Observation* (The locus of the math-physics gap). Combining [1, §16.5] (no CM-internal source for  $(2, 3)$ ) with Observation 4 (joint match uniquely fixes  $(2, 3)$ ): the gap between  $\chi_{-4}$ -derived mathematical structure and the FTD physical identification is localised precisely at the polynomial *assembly* step. The integer coefficient 16 and the constant  $G^*$  are both forced by  $\chi_{-4}$ . The exponent pair  $(2, 3)$  is forced only by the joint physical-matching requirement. Closing this last gap — if a clean derivation of  $(2, 3)$  from  $\chi_{-4}$ -internal data exists — would convert the FTD bridge from a two-arrow conjecture into a single-arrow theorem.

## 5. THE FULL DESCENT CHAIN

The cumulative content of [1] and this companion can be displayed as a single descent chain from physics to ontological bedrock:

$$\begin{array}{c}
 \underbrace{(\alpha^{-1}, N_c)}_{\text{measured physical constants}} \\
 \uparrow \quad \text{[STRONGLY MOTIVATED CONJECTURE] (FTD bridge)} \\
 \underbrace{P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}}_{\text{master quadratic}} \\
 \uparrow \quad (2, 3) \text{ exponents: uniquely forced by joint root match [1, §16.6]} \\
 \underbrace{\{16, G^*\}}_{\text{building blocks}} \\
 \uparrow \quad \text{four functorial projections [1, Thm 16.2]} \\
 \underbrace{\chi_{-4}}_{\text{Kronecker character of } \mathbb{Q}(i)} \\
 \uparrow \quad \text{imaginary-part trace } \chi_{-4}(n) = \text{Im}(i^n) = \sin(\pi n/2) \\
 \underbrace{\{-1, 0, +1\}}_{\text{value set = FTD voxel alphabet (set-level)}} \\
 \uparrow \quad \text{closure of } \langle i \rangle\text{-orbit under squaring \& modulus} \\
 \underbrace{i^2 = -1}_{\text{ontological zero-point}}
 \end{array}$$

Every arrow above the FTD bridge is established (theorem, observation, or honest conjecture) in [1]. The FTD bridge itself is the physical conjecture that this companion records under [STRONGLY MOTIVATED CONJECTURE]. Everything below the bridge is pure mathematics; the only place that requires physical input is the single bit of the exponent-pair selector.

## 6. OUTLOOK

Two productive lines of work are identified by the descent chain.

**Closing the bridge.** If a derivation of the exponent pair  $(2, 3)$  from  $\chi_{-4}$ -internal data can be made rigorous — the candidate framing in [1, §16.5, Conjecture (Sym<sup>2</sup>–Sym<sup>3</sup>)] is the most natural starting point — the FTD bridge would become a single-arrow theorem rather than a two-arrow conjecture. This would not prove the empirical match to  $(\alpha^{-1}, N_c)$  (that remains an experimental fact about the universe), but it would localise the conjectural content to a single physical assertion: that the universe’s electroweak and strong sectors are built from a  $\mathbb{Z}[i]$ -substrate.

**Extending the bridge.** The current FTD identifications cover  $\alpha$  and  $N_c$ . The Standard Model has additional dimensionless constants ( $\sin^2 \theta_W$ ,  $m_\mu/m_e$ ,  $m_\tau/m_e$ , PMNS angles, etc.) which the FTD framework currently parameterises without first-principles derivation. The  $\chi_{-4}$ -unification suggests that further dimensionless invariants may be discoverable as roots of higher-rank master polynomials living in  $\text{Sym}^\bullet(H^1(E_{\text{lemn}}))$ , or as Hecke L-values of related CM elliptic curves. This is the natural program for extending the bridge from  $(\alpha^{-1}, N_c)$  to the full SM.

A second-tier program suggested by the polymath synthesis of [1, appendix] is the identification of the  $\chi_{-3}$ -side (equianharmonic,  $|\text{Aut}| = 6$ ) as the structural home of the QCD gauge structure, with  $\chi_{-4}$  (lemniscatic) as the home of the electroweak structure. The Moore Layer Theorem of FTD [2] produces  $U(1) \times SU(2) \times SU(3)$  from a 27-block polyhedral decomposition; whether this can be reformulated as the joint action of  $\chi_{-4}$  (giving  $U(1)$  and part of  $SU(2)$ ) and  $\chi_{-3}$  (giving

$SU(3)$  and the rest of  $SU(2)$  via the Hurwitz-quaternion intersection  $\mathbb{H} \supset \mathbb{Z}[i], \mathbb{Z}[\rho]$ ) is an open speculative direction worth investigating but not pursued here.

## REFERENCES

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